

Lecture 2

mathematical preliminaries

Landau's big- $\mathcal{O}$ order symbol

Uses of big- $\mathcal{O}$ notation

In Lecture 1 we had already used the $\mathcal{O}$ -symbol:

$$\begin{aligned}\text{fl}(x_1 + x_2) &= \text{fl}(\text{fl}(x_1) + \text{fl}(x_2)) = (x_1(1 + \epsilon_1) + x_2(1 + \epsilon_2))(1 + \epsilon_{c2}) \\ &= x_1 + x_2 + x_1(\epsilon_1 + \epsilon_{c2}) + x_2(\epsilon_2 + \epsilon_{c2}) + x_1\epsilon_1\epsilon_{c2} + x_2\epsilon_2\epsilon_{c2} \\ &= x_1 + x_2 + x_1(\epsilon_1 + \epsilon_{c2}) + x_2(\epsilon_2 + \epsilon_{c2}) + \mathcal{O}(\epsilon^2)\end{aligned}$$

Definition: $\mathcal{O}$ -symbol

Let f and g be given functions. Then,

$$f(x) = \mathcal{O}(g(x)), \quad \text{for } x \rightarrow 0,$$

if there exists $r > 0$ and $M < \infty$ such that

$$|f(x)| \leq M|g(x)|, \quad \text{for all } x \in [-r, r]$$

Big- $\mathcal{O}$ notation: example

Let $f(\epsilon) = x_0 + x_1\epsilon + x_2\epsilon^2 + x_3\epsilon^3$. Then,

$$f(\epsilon) = x_0 + x_1\epsilon + \mathcal{O}(\epsilon^2), \quad \text{for } \epsilon \rightarrow 0.$$

Proof. We claim that $x_2\epsilon^2 + x_3\epsilon^3 = \mathcal{O}(\epsilon^2)$ for $\epsilon \rightarrow 0$, i.e., that there exist $r > 0$ and $M < \infty$, such that

$$|x_2\epsilon^2 + x_3\epsilon^3| \leq M|\epsilon^2|, \quad \text{for all } \epsilon \in [-r, r].$$

Take any $0 < r < \infty$. For all $\epsilon \in [-r, r]$, it holds that $|\epsilon| \leq r$ and

$$\begin{aligned} |x_2\epsilon^2 + x_3\epsilon^3| &= |x_2 + x_3\epsilon| |\epsilon^2| \leq (|x_2| + |x_3| |\epsilon|) |\epsilon^2| \\ &\leq (|x_2| + |x_3|r) |\epsilon^2| = M|\epsilon^2|, \end{aligned}$$

where $M = |x_2| + |x_3|r < \infty$.

Computations with $\mathcal{O}$ symbol

Let $f(x) = \mathcal{O}(x^p)$ and $g(x) = \mathcal{O}(x^q)$ for $x \rightarrow 0$ with $p, q \geq 0$. Then,

$$f(x) = \mathcal{O}(x^p) = \mathcal{O}(x^s) \quad \text{for all } 0 \leq s \leq p$$

$$\alpha f(x) + \beta g(x) = \alpha \mathcal{O}(x^p) + \beta \mathcal{O}(x^q) = \mathcal{O}(x^{\min(p,q)}) \quad \text{for all } \alpha, \beta \in \mathbb{R}$$

$$f(x)g(x) = \mathcal{O}(x^p)\mathcal{O}(x^q) = \mathcal{O}(x^{p+q})$$

$$\frac{f(x)}{|x|^s} = \frac{\mathcal{O}(x^p)}{|x|^s} = \mathcal{O}(x^{p-s}) \quad \text{if } 0 \leq s \leq p.$$

Computations with $\mathcal{O}$ symbol: example

Let $f(x) = \mathcal{O}(x^3)$ and $g(x) = \mathcal{O}(x^2)$ for $x \rightarrow 0$. Then,

$$f(x) = \mathcal{O}(x^3), \quad \text{but also} \quad f(x) = \mathcal{O}(x^2) \quad \text{and} \quad f(x) = \mathcal{O}(x)$$

$$3\mathcal{O}(x^3) - 2\mathcal{O}(x^2) = \mathcal{O}(x^2)$$

$$\mathcal{O}(x^3)\mathcal{O}(x^2) = \mathcal{O}(x^5)$$

$$\frac{\mathcal{O}(x^3)}{|x|^2} = \mathcal{O}(x)$$

Concepts and theorems from analysis



Jacques Hadamard (1865–1963)

Definition: well-posed and ill-posed problems

A mathematical problem is *well-posed* if it satisfies the following conditions:

- 1 A solution exists
- 2 The solution is unique
- 3 The solution changes continuously with the data

If a problem fails to satisfy any of these conditions, then it is *ill-posed*.

Not all well-posed problems are equally good.

Definition: well-conditioned and ill-conditioned problems/models

A problem/model is called *well-conditioned* if small errors in the data lead to small errors in the results. Otherwise the problem/model is *ill-conditioned*.

One problem can be better conditioned than the other.

Related concept: *condition number* of a matrix.

Definition: modulus of a complex number

Let $z \in \mathbb{C}$, i.e., $z = a + ib$, $a, b \in \mathbb{R}$. Then,

$$|z| = |a + ib| = \sqrt{a^2 + b^2}.$$

Let $\bar{z} = a - ib$ be the complex conjugate of z . Then,

$$\bar{z}z = (a - ib)(a + ib) = a^2 + b^2 = |z|^2.$$

However,

$$zz = z^2 = (a + ib)(a + ib) = a^2 - b^2 + i2ab \neq |z|^2$$



Leopold Kronecker (1823–1891)

Definition: Kronecker's delta

$$\delta_{i,j} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Examples:

$$\int_0^\pi \sin(m\phi) \sin(n\phi) d\phi = \frac{\pi}{2} \delta_{m,n} (1 - \delta_{n,0})$$

$$\sum_{j=1}^n a_{i,j} \delta_{j,k} = a_{i,k}, \quad \text{in particular,} \quad \sum_{j=1}^n a_{i,j} \delta_{j,i} = a_{i,i}$$

Definition: eigenvalues and eigenvectors

Let $A \in \mathbb{C}^{m \times m}$ be any matrix and $I \in \mathbb{R}^{m \times m}$ - the identity matrix. Then, the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$ of A satisfy:

$$\det(A - \lambda I) = 0,$$

and

$$A\mathbf{v}_i = \lambda_i \mathbf{v}_i,$$

where $\mathbf{v}_i \in \mathbb{C}^m$ is the eigenvector corresponding to the eigenvalue λ_i .

Eigen-decomposition

With $A \in \mathbb{R}^{n \times n}$, let $V = [\mathbf{v}_1, \dots, \mathbf{v}_n] \in \mathbb{R}^{n \times n}$ be the matrix with the eigenvectors $\mathbf{v}_i$ as columns, and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n) \in \mathbb{R}^{n \times n}$ - the diagonal matrix with the eigenvalues λ_i on its diagonal. Then,

$$A\mathbf{v}_i = \lambda_i\mathbf{v}_i \Leftrightarrow AV = V\Lambda.$$

If V^{-1} exists, then there exists the *eigen-decomposition* of A :

$$A = V\Lambda V^{-1},$$

and the matrix A can be *diagonalized* by V :

$$V^{-1}AV = \Lambda$$

In general, even if all entries $[A]_{i,j}$ of A are real, i.e., $A \in \mathbb{R}^{n \times n}$, the eigenvalues λ_i of A can be complex.

Eigen-decomposition of symmetric matrices

A real *symmetric matrix* $A \in \mathbb{R}^{n \times n}$ is such that $A^T = A$, i.e., $[A]_{i,j} = [A]_{j,i}$. Among many other, it has the following properties:

- Every real symmetric matrix can be diagonalized by an orthogonal matrix V :

$$V^T A V = \Lambda, \quad \text{with} \quad V^T V = I.$$

- The eigenvectors are mutually orthogonal, i.e., $\mathbf{v}_i^T \mathbf{v}_j = \delta_{i,j}$
- The eigenvalues are real, $\lambda_i \in \mathbb{R}$, $i = 1, \dots, n$.

Triangle inequalities

For any $x, y \in \mathbb{R}$:

$$|x + y| \leq |x| + |y|,$$
$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

Useful for computing upper bounds. Example of application:

$$|f(x) - g(x)| = |f(x) - v(x) + v(x) - g(x)| \leq |f(x) - v(x)| + |v(x) - g(x)|$$

Definition: sets of continuous functions

The set of all continuous functions on the interval $[a, b]$ is denoted as $C[a, b]$. The set of all functions whose p -th derivative is continuous on $[a, b]$ is denoted as $C^p[a, b]$.

Function $f(x)$ is differentiable at x , if its derivative $f'(x)$ exists at x .

All differentiable functions are continuous, and $C^1[a, b] \subset C[a, b]$.

Not all continuous functions are differentiable.

Not all differentiable functions have a continuous derivative.

Theorem: intermediate value

Let $f \in C[a, b]$ and $f(a) \neq f(b)$. For any number F between $f(a)$ and $f(b)$ there exists a number $c \in (a, b)$ such that $f(c) = F$.

Application: consider $f(x_1) \neq f(x_2)$ and the average:

$$\frac{f(x_1) + f(x_2)}{2}$$

It is clear that the number $F = (f(x_1) + f(x_2))/2$ is between $f(x_1)$ and $f(x_2)$. Hence, from the Intermediate Value Theorem, there is a point ξ such that

$$\frac{f(x_1) + f(x_2)}{2} = f(\xi)$$



Michel Rolle (1652–1719)

Theorem (Rolle):

Let $f \in C[a, b]$ and $f(x)$ is differentiable on (a, b) . If $f(a) = f(b)$, then there exists a number $c \in (a, b)$ such that $f'(c) = 0$.

Application: Rolle's Theorem helps finding extrema of a function.

Theorem: mean value

Let $f \in C[a, b]$ and $f(x)$ is differentiable on (a, b) . Then, there exists a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Theorem: mean value for integrals

Let $g \in C[a, b]$ and $f(x)$ is integrable and does not change sign on $[a, b]$. Then, there exists a number $c \in (a, b)$ such that

$$\int_a^b g(x)f(x) dx = g(c) \int_a^b f(x) dx$$



Weierstrass

Karl Weierstrass (1815–1897)

Theorem (Weierstrass): uniform polynomial approximation

Let $f \in C[a, b]$. For every $\epsilon > 0$ there exists a polynomial function $P(x)$ such that:

$$|f(x) - P(x)| < \epsilon, \quad \text{for all } x \in [a, b]$$



Brook Taylor (1685–1731)

Theorem: Taylor's polynomial

Let $f \in C^{n+1}[a, b]$. Then, for all $c, x \in (a, b)$ there exists a number ξ between x and c such that:

$$f(x) = P_n(x) + R_n(x),$$

where

$$P_n(x) = f(c) + f'(c)(x - c) + \frac{f''(c)}{2!}(x - c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x - c)^n$$

is the *Taylor polynomial* and

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - c)^{n+1}$$

is the *remainder*.

Taylor series

We shall often deal with the values of $f(x)$ in neighboring points $f(x - 2h)$, $f(x - h)$, $f(x + h)$, $f(x + 2h)$, etc. These can be related to f and its derivatives at x via the Taylor series. We use the Taylor polynomial with $c = x$ and with $x + h$, $x - h$, etc., instead of x , so that $x - c$ becomes $x + h - x = h$ or $x - h - x = -h$, etc.

$$f(x + h) = f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \frac{f'''(x)}{6}h^3 + \frac{f^{(4)}(\xi)}{24}h^4, \quad \xi \in (x, x + h)$$

$$f(x - 2h) = f(x) - 2f'(x)h + 2f''(x)h^2 - \frac{4f'''(x)}{3}h^3 + \frac{2f^{(4)}(\xi)}{3}h^4, \quad \xi \in (x - h, x)$$

We can also write:

$$f(x + h) = f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \frac{f'''(x)}{6}h^3 + \frac{f^{(4)}(x)}{24}h^4 + \mathcal{O}(h^5)$$